

Homework 3

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2026-05-28

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Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects)

# storing kelp data as object "kelp"
kelp <- read_csv(here("data", "temp-kelp.csv"))
# creating new object that fits a linear model to the data
kelp_model <- lm(
  kelp_elong ~ temp_c,
  data = kelp)
# storing personal data as object "personal_data"
personal_data <- read_xlsx(here("data", "ENVS 193DS- Personal Data.xlsx"))
```

Problem 1

1a.

The appropriate tests would be the Pearson's correlation and linear regression to determine the strength of the relationship between giant kelp frond elongation rate and temperature. Pearson's correlation measures the strength and the direction between the two variables. Linear regression aims to describe a linear relationship between the variables while being able to describe strength as well.

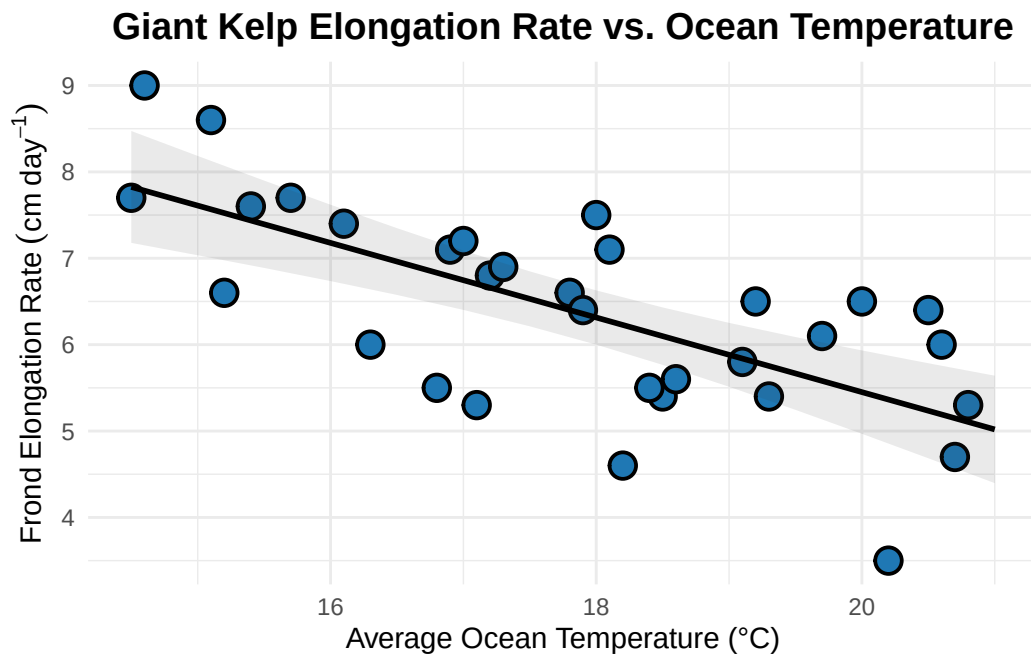
1b.

```
# creating new object called kelp_preds
kelp_preds <- ggpredict(
  kelp_model,
  terms = "temp_c")
# base layer: ggplot with kelp data
ggplot(data = kelp,
  aes (x = temp_c, # x-axis is sea temp
        y = kelp_elong)) + # y-axis is kelp elongation rate
# first layer: geom_point to show underlying data
geom_point(fill = "#1f78b4", # changing the color and size
  size = 4,
  stroke = 1,
  shape = 21) +
# second layer: ribbon representing confidence interval
# using predictions data frame
geom_ribbon(data = kelp_preds,
  aes(x = x,
        y = predicted,
        ymin = conf.low,
        ymax = conf.high),
  alpha = 0.1) +
# third layer: line representing model predictions
geom_line(data = kelp_preds,
  aes(x = x,
        y = predicted),
  linewidth = 1)+
labs ( x= "Average Ocean Temperature (°C)",
  y = bquote(Frond~Elongation~Rate~(cm~day^-1)),
```

```

    title = "Giant Kelp Elongation Rate vs. Ocean Temperature") +
# changing the theme
theme_minimal() +
# customizing the look of the titles!
theme(plot.title = element_text(face = "bold",
                                size = 14,
                                hjust = 0.5),
      axis.title = element_text(face = "plain",
                                size = 11))

```



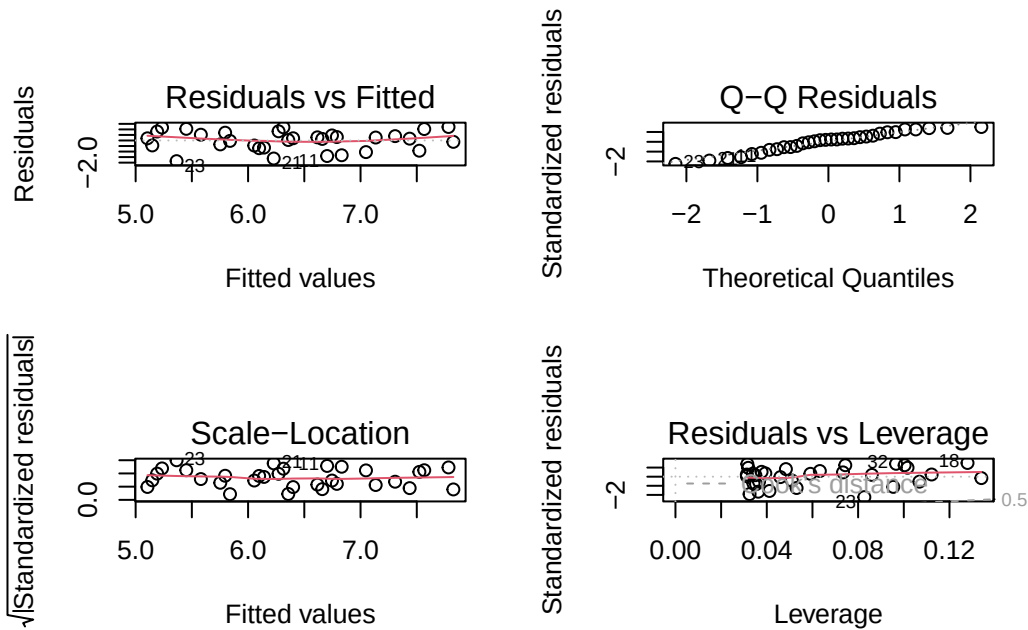
1c.

Checking assumptions

```

# base R residuals
par(mfrow = c(2,2))
plot(kelp_model)

```



When thinking about the relationship between ocean temperature and kelp elongation, it would make sense for there to be a linear relationship. We can see this in the residuals vs. fitted plot and the plotted data. We can assume independent errors due to the independent observations and data collection methods. The residual vs. fitted and scale location plots both show us the homoscedasticity of errors. The plots show that the data is homoscedastic because of the decently straight trend line. Looking at the QQ plot seen in residual plots, the data seems to be normally distributed. It roughly follows the straight line with variations that are normal to see in real data.

I think I did ok checking the assumptions. It's hard because making the decision is subjective, but generally most math isn't, and making the "wrong" choice can have big impacts. The QQ plot was hardest to assess due to the slight curve seen in it.

Running test

```
# showing the output of linear regression
summary(kelp_model)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8643	-0.5017	0.1790	0.5659	1.2177

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.08645	1.49219	9.440	1.72e-10 ***
temp_c	-0.43179	0.08321	-5.189	1.37e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 1 and 30 DF, p-value: 1.366e-05